had antibacterial properties. Indeed, traditional remedies for infection had included moldy bread or fruit. When Fleming published his findings in 1929, initial prospects for exploiting the antibiotic were mixed: the mold was difficult to cultivate, the quantities of penicillin produced were tiny, and it could not be stored for long. As a result, Fleming effectively lost interest and gave up on the project for the next decade.

Fortunately, in 1939, there was another breakthrough: a team at Oxford University headed by Howard Florey (see box) came across Fleming's research and were able to radically improve the cultivation, extraction, and purification processes for penicillin. In 1941, the antibiotic was trialed on a man named Albert Alexander, who had abscesses and blood poisoning. Injections of penicillin had a dramatic effect, but the supplies ran out, and sadly Albert died. However, four of the next five patients recovered from their infections. Drug companies became interested and invested in developing higher-yielding strains of *Penicillium*. Finally, penicillin could be made on a large scale. It proved highly effective against a wide range of bacterial infections, including scarlet fever, meningitis, and pneumonia.

What had once been unimaginable was now possible: bacterial infection could be effectively treated. The landscape of medicine had changed.

HOWARD **FLOREY**

**Australian Florey led the team who learned how to produce penicillin in large quantities.**

Florey (1898–1968), a pathologist at Oxford University, was researching natural antibacterials with biochemist Ernst Chain. Aided by Norman Heatley, they developed the expertise to manufacture and purify penicillin. Using all kinds of equipment, and with funding from the US, they made sufficient quantities of penicillin to test it first on mice and then on a human. Fleming and Florey were both knighted for their work.

IN **WORLD WAR II,** **TWO-THIRDS** OF **SOLDIERS** WITH **CHEST WOUNDS** WERE **SAVED BY PENICILLIN**

**10 DOSES** **OF PENICILLIN** WERE MADE IN **1942;** **600 BILLION** DOSES IN **1945**